Reinforcement Learning with Verifiable Rewards Implicitly Incentivizes Correct Reasoning in Base LLMs

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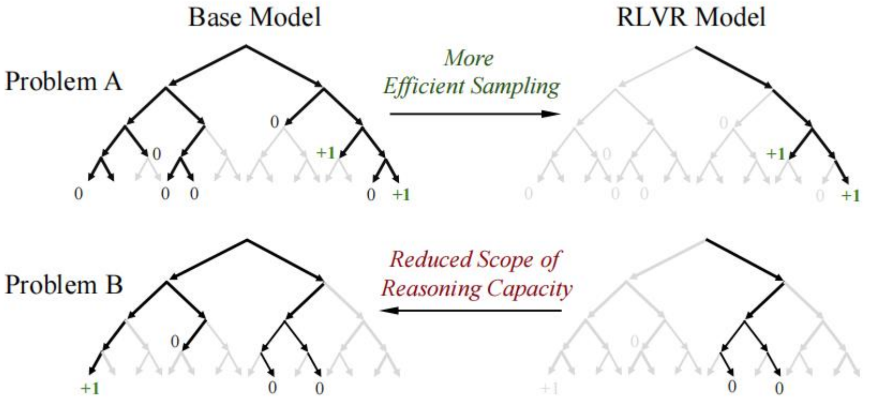
MSRA
ICLR2026

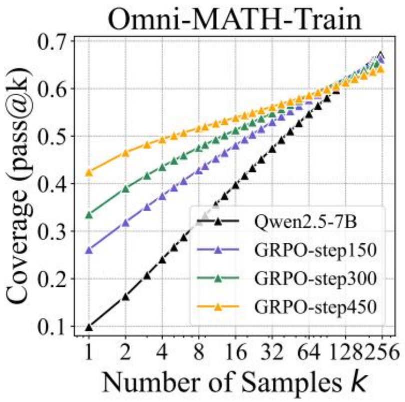
RLVR能否拓展LLM能力边界?

反方：（NIPS2025）Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

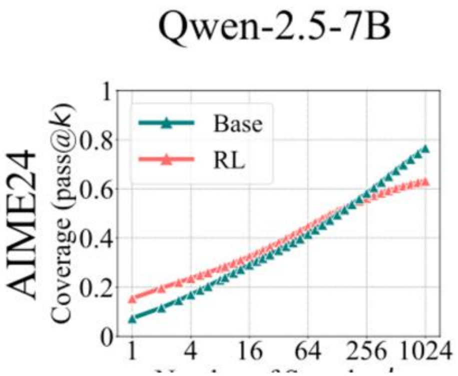
用pass@k衡量能力边界：给模型K次机会，只要有一次做对就算通过。k越大，越能探测模型“理论上能不能做对”的上限。

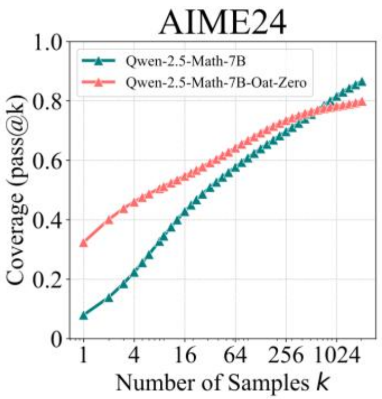
结论：RLVR只是在base model已有的推理路径上做了更高效的采样，牺牲了推理多样性，没有拓展能力边界





- K比较小的时候，RL > Base：RLVR 能提升采样到正确答案的概率。
- 随着K增大，base的 pass@k 曲线上升斜率明显大于 RL 模型
- 足够大的 k 值下，Base > RL：base模型在可解问题上的覆盖范围更广





RLVR能否拓展LLM能力边界?

正方：（ICLR2026）REINFORCEMENT LEARNING WITH VERIFIABLE RE_x0002_WARDS IMPLICITLY INCENTIVIZES CORRECT REASONING IN BASE LLMS

pass@k：只检查最终答案，而不检查推理过程（CoT）是否正确！base model的输出分布更分散，更容易蒙对

CoT-Pass@k：只有当答案正确且CoT也正确，才算正确。用 DeepSeek-R1-0528-Qwen3-8B 作为 CoT 验证器，对每条推理链做 3 次独立验证

实验结果：在pass@k上复现反方结论。在CoT-Pass@k上（特别是AIME25），RL始终大于Base；Math500和AMC23太简单，差异不明显；Minerva可能因为数据域不匹配，RL提升有限。

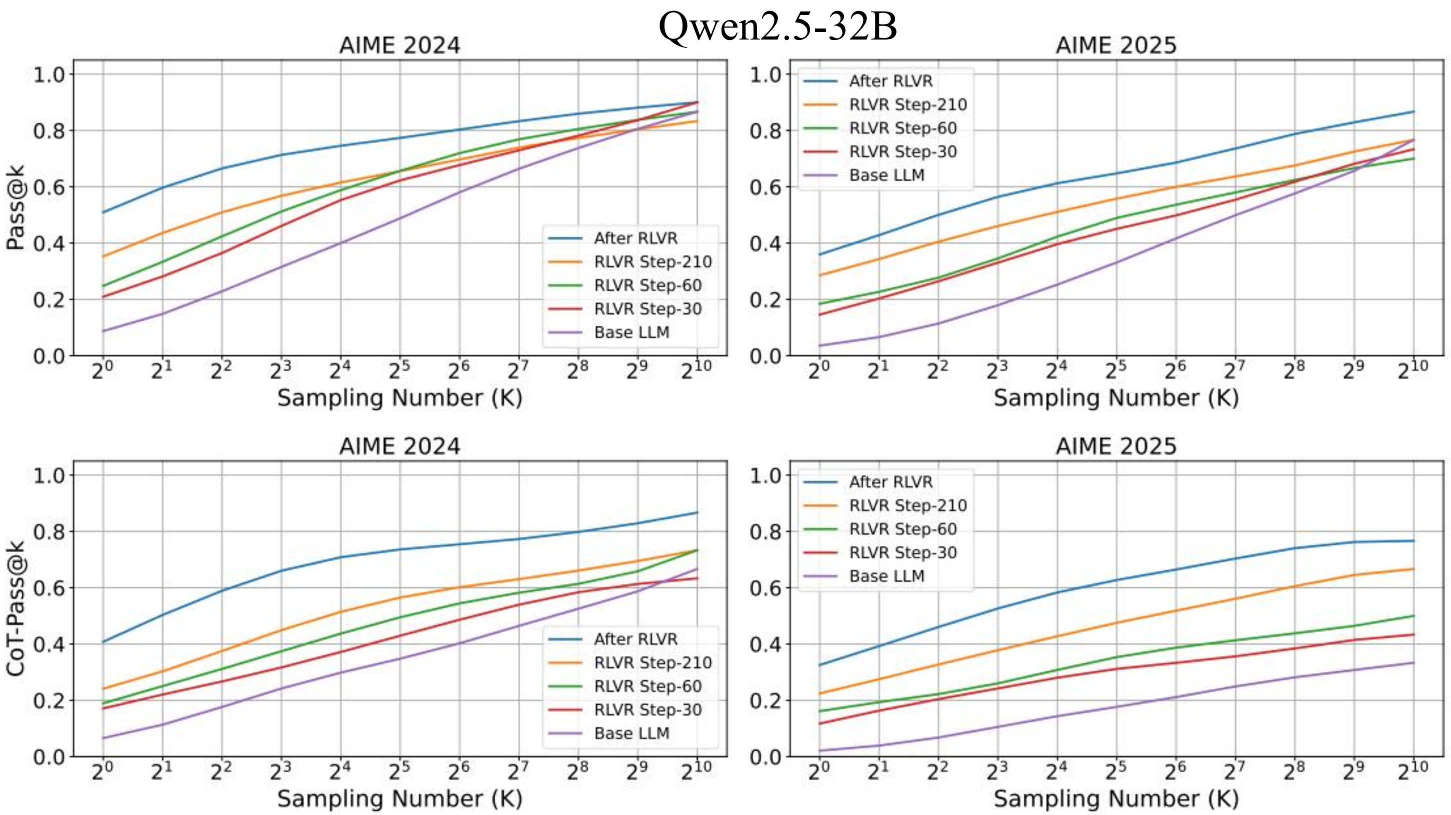
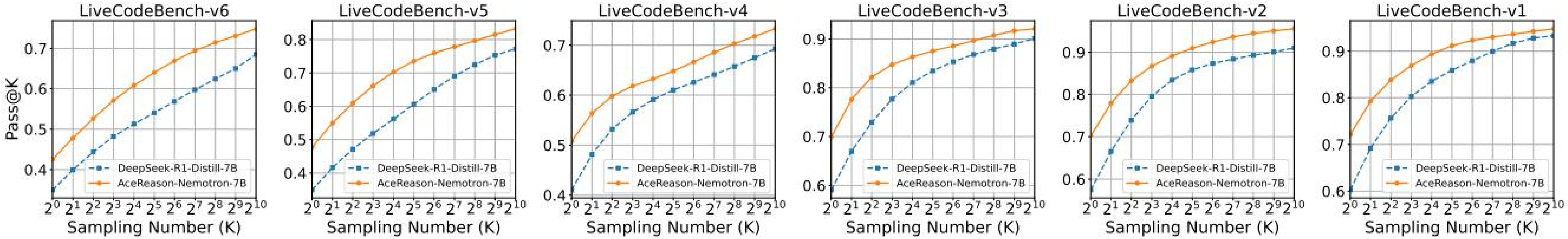


Figure 5: The evolution of Pass@K (the top row) and CoT-Pass@K (the bottom row) performance on AIME 2024 and 2025 for different model checkpoints during the DAPO training.

RLVR能否拓展LLM能力边界?

正方：（ICLR2026）REINFORCEMENT LEARNING WITH VERIFIABLE RE_x0002_WARDS IMPLICITLY INCENTIVIZES CORRECT REASONING IN BASE LLMS

在code领域，pass@k可作为CoT-Pass@k的有效代理，在更难的基准上，RL模型和原模型（蒸馏模型）差异更明显。



反方：基于base模型的RL，实验结果支持反方结论（左）。 基于Distill模型的RL，实验结果与正方类似（右）

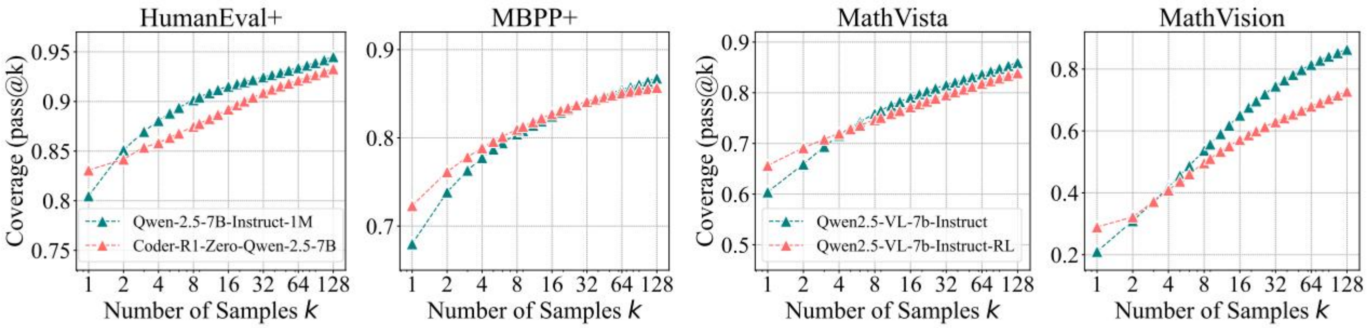


Figure 4: Pass@k curves of base and RLVR models. **(Left)** Code Generation. **(Right)** Visual Reasoning.

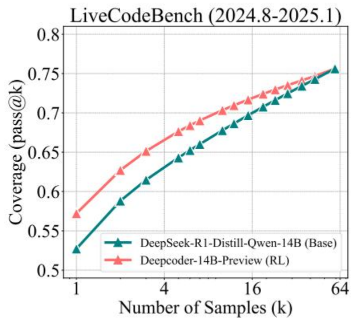


Figure 3: RLVR for Coding.

理论解释&训练验证：为什么只验证答案对错也能激励正确推理？

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核心先验假设：正确CoT比错误的CoT有更高概率得到正确答案

$$P(\text{答案正确} \mid \text{CoT正确}) = \alpha > P(\text{答案正确} \mid \text{CoT错误}) = \beta。$$

在这个 假设下， $\mathbb{E}[\hat{A}(y_i) \mid \text{CoT正确}] > 0$ 而 $\mathbb{E}[\hat{A}(y_i) \mid \text{CoT错误}] < 0$。

使得GRPO能通过 α-β 这个信号隐式增加正确CoT产生的概率

训练验证：在DAPO训练集上，答案正确率P(CA)持续上升，P(CoT正确|答案正确) 亦持续上升.
存在瓶颈：即使P(CA)接近1.0，P(CC|CA)仍只有0.7，即只依靠答案正确作为reward信号，推理质量提升有上限

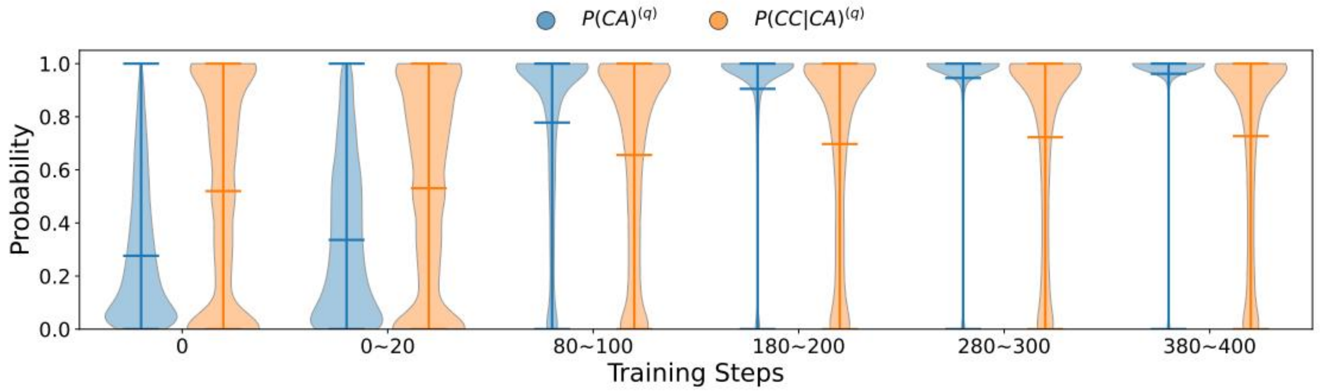


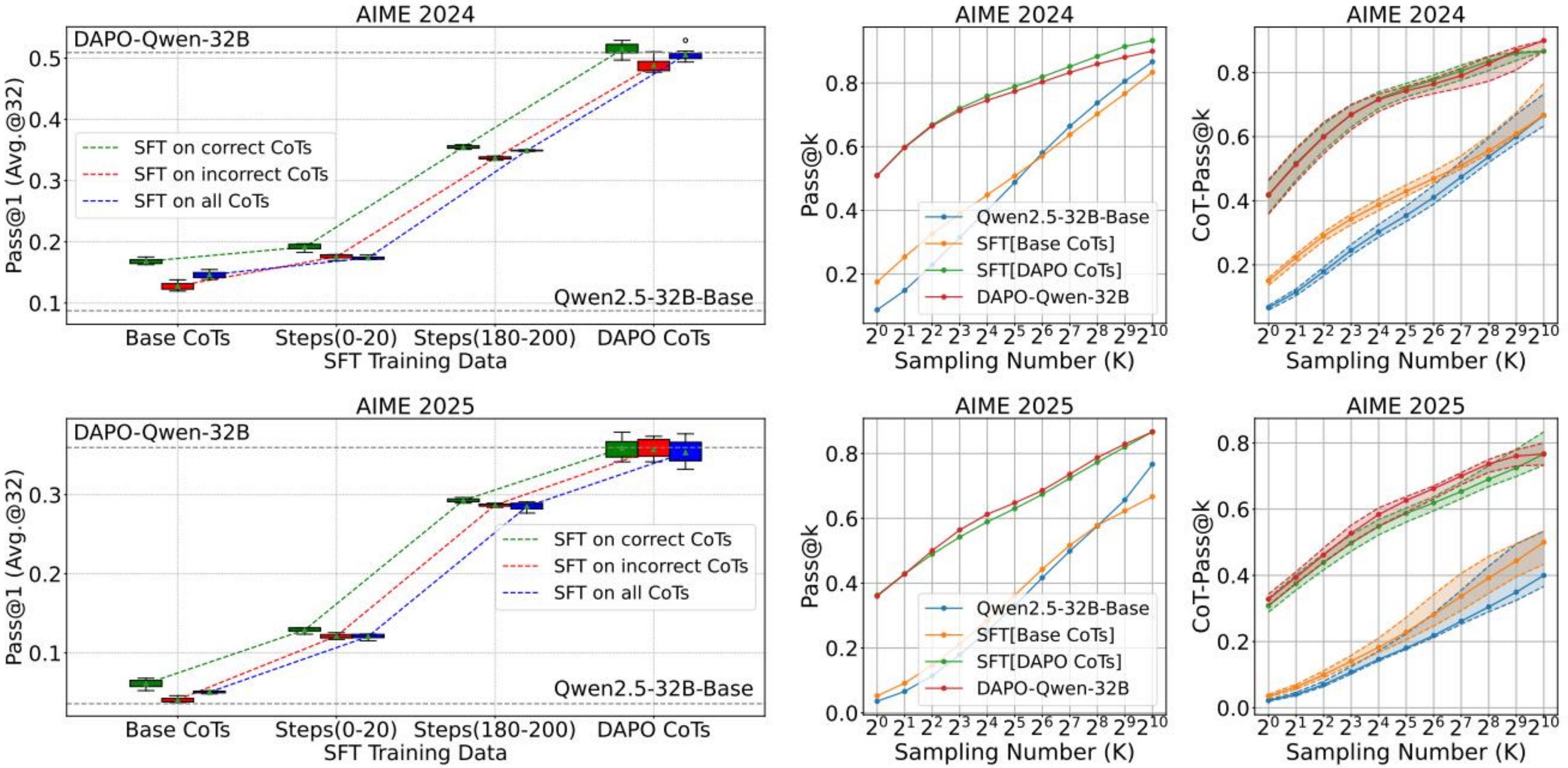
Figure 4: The evolution of $P(CA)^{(q)}$ (the fraction of correct answers for prompt q) and $P(CC|CA)^{(q)}$ (the fraction of correct CoTs within the correct answers for prompt q) for fully optimized training questions over the course of DAPO training.

RL后模型的CoT质量分析

左：使用不同RL阶段的CoT数据对base模型进行SFT。随着RL的进行，CoT质量不断提升，SFT后的模型性能也不断提升，直至复原DAPO-Qwen-32B能力

右：使用不同CoT数据对base模型进行SFT。

说明RLVR提供了新的高质量的CoT数据，而这些是base模型难以直接采样得到的



(a) The CoT quality at different RLVR stages, using Pass@1 on test sets as the proxy metric.

(b) The CoT quality before and after RLVR, using (CoT-)Pass@K on test sets as the proxy metric.

ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models

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NIPS2025

长期RL训练方案（超2000steps）

长期RL训练中存在两大挑战：熵崩溃 和 训练不稳定

方案优化：GRPO+ 动态采样（DAPO）+ clip上下限解耦（DAPO）+ KL约束

关于KL使用的观点：近期研究主张移除KL约束，因为模型在推理训练中会自然产生发散现象。本文认为这一主张适用于没有SFT的base模型。而本文使用的DeepSeek-R1-Distill-Qwen-1.5B已具有良好推理能力。

2.1 Background: Group Relative Policy Optimization

We adopt Group Relative Policy Optimization (GRPO) [16] as the core RL algorithm. Compared with Proximal Policy Optimization (PPO) [17], it removes the value model and instead use baseline estimates based on group scores. Formally the GRPO maximizes the following objective:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\min\Big(r_\theta(\tau)A(\tau),\quad \text{clip}(r_\theta(\tau), 1-\epsilon, 1+\epsilon)A(\tau)\Big)\right], \tag{1}$$

where τ is the response sampled from the current policy π_θ. $r_\theta(\tau) = \frac{\pi_\theta(\tau)}{\pi_{old}(\tau)}$ is the probability ratio between the current policy and old policy before each actor update. The advantage used in GRPO foregoese the critic model of PPO, and instead estimates baseline from group scores $\{R_i\}_{i\in G(\tau)}$:

$$A(\tau) = \frac{R_\tau - mean(\{R_i\}_{i\in G(\tau)})}{std(\{R_i\}_{i\in G(\tau)})}. \tag{2}$$

$$\text{clip}(r_\theta(\tau), 1-\epsilon_{low}, 1+\epsilon_{high}).$$

$$L_{KL-RL}(\theta) = L_{GRPO}(\theta) - \beta D_{KL}(\pi_\theta||\pi_{ref}).$$

实验结果

长期上，在各个基准上都遥遥领先训练起点DeepSeek-R1-Distill-Qwen-1.5B，证明长期RL是有收益的，同时也没有熵崩溃现象。

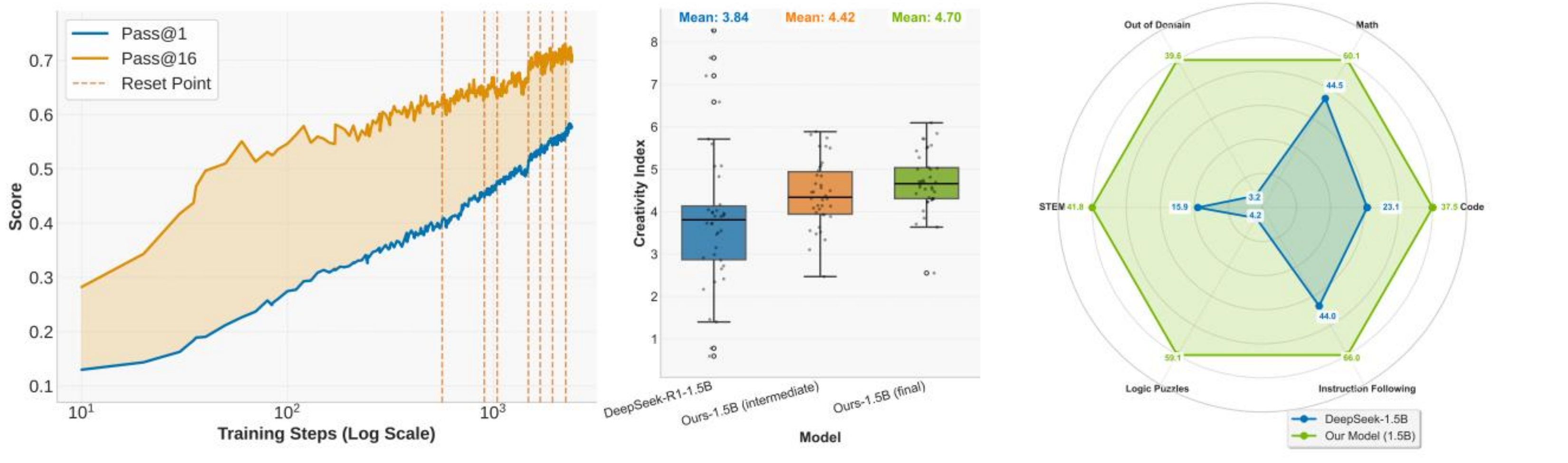


Figure 1: Benefits of prolonged reinforcement learning (ProRL). **Left**: Pass@1 and Pass@16 scales with ProRL training. **Middle**: ProRL leads to more novel solutions reflected by higher Creativity Index [12]. **Right**: Our model greatly surpass base model across diverse tasks.

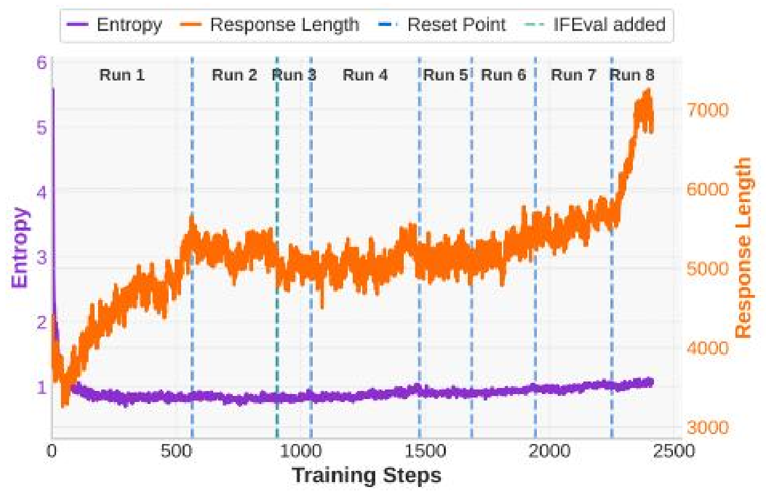


Figure 2: ProRL training dynamics.

关于RL与LLM能力边界？ProRL多领域测试结果

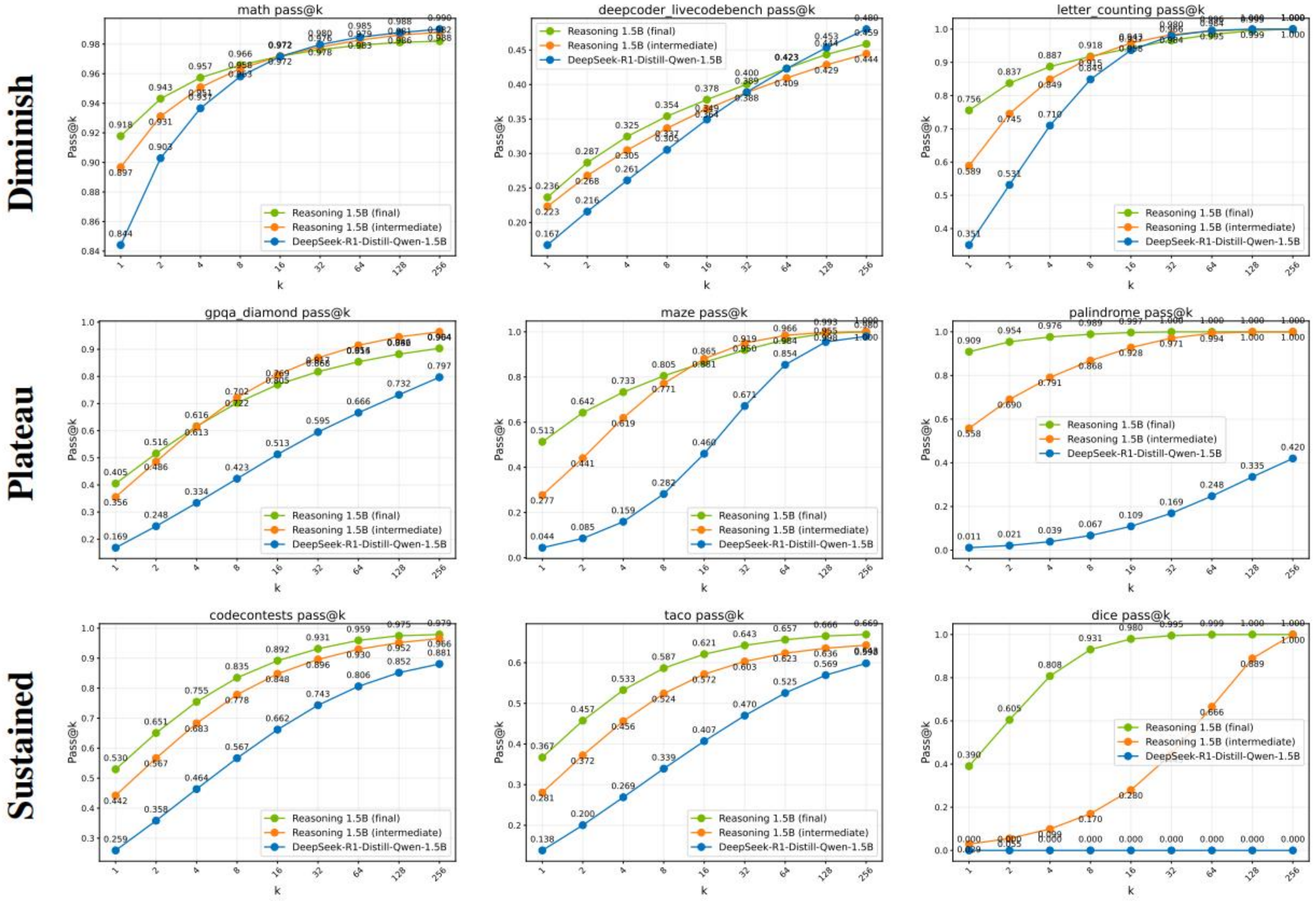
先前研究数据域太窄了，RL训的step太少了！

有三种情况：

1. diminish：pass@1提升，但pass@128下降。这类任务通常base模型具有较高的pass@128，表明base模型本身已具备足够推理能力，RL只是在优化采样效率

2.Plateau：超长RL训练时，模型性能主要出现在训练初期（对比橙和绿），ProRL带来的收益微弱，说明模型在这些任务上迅速达到收敛状态。

3.Sustained：某些测试上模型能力持续提升，甚至在base模型完全不会做的任务上达到了100%的准确率



关于RL与LLM能力边界？ base越弱的领域，RL带来的提升越大

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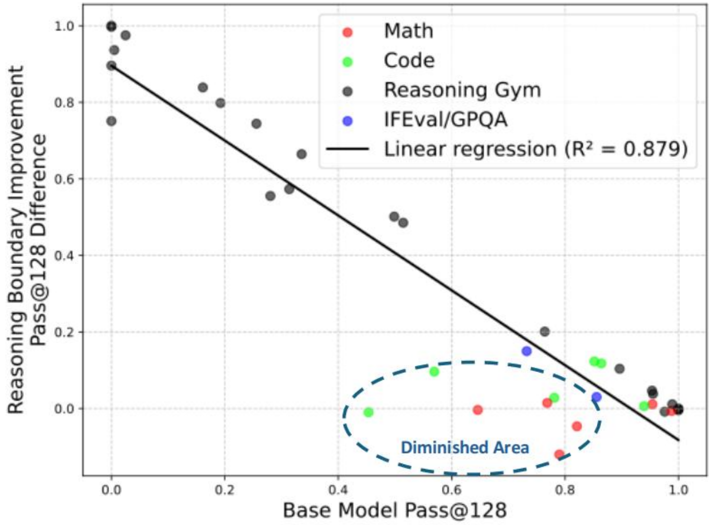
base模型越弱的领域，RL带来的提升越大；
base模型越强的领域，RL带来的提升越小，甚至有负收益



OOD表现：
1. 左：训练集中不包含的任务boxnet。
ProRL涌现了解题能力

2. 右：训练集中只包含1~10size的题目，
在10~20size上进行OOD测试（任务难度增加），ProRL仍然有提升

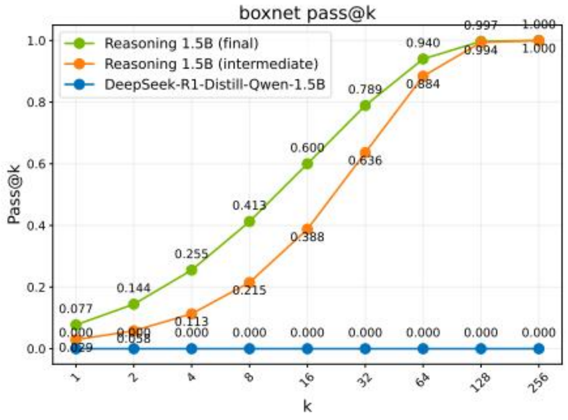


Figure 5: Expanded reasoning boundary for OOD task *boxnet*.

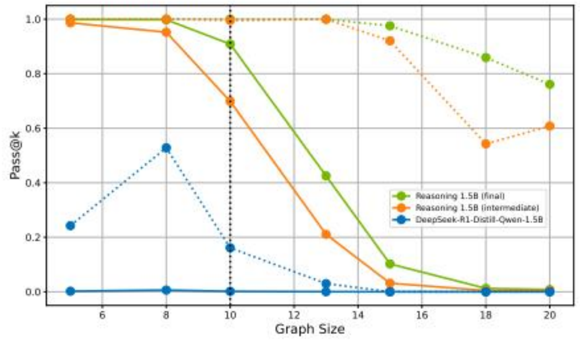


Figure 6: ProRL generalizes to increased task difficulty on *graph_color*.

On the Interplay of Pre-Training, Mid-Training, and RL on Reasoning Language Models

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arixiv 2025年12月

Introduction

近期研究争论RL能否提升LLM能力边界，但是这些研究基于不可控的训练环境，无法确定模型学会了哪些基本推理能力。

本文构建了一个可控训练环境，独立评估每个训练阶段对模型的贡献，包括pre-training、mid-training、post-training。

核心发现：

- RL只有在满足两个条件时才能提升能力边界：任务在预训练阶段未充分学习；RL数据需匹配模型能力边界（不能太难也不能太简单）
- RL只在base模型具备一些base skills时，才能实现情景泛化。
- mid-training阶段能显著提升ID和OOD表现，是值得重视的训练阶段
- process reward可抑制reward hacking并提升推理一致性，可在复杂组合场景下提升模型能力

可控训练环境构造

训练数据：通过有向无环图、情景模板生成多种数学题。
有向无环图：节点代表数字、边代表运算。题目难度通过运算次数（op）控制
情景模板：包装题目场景，如动物园、学校、公园…

任务设定：
1. 外推泛化，例如预训练使用op=2~10的题目，测试使用op=11~14、15~20的题目
2. 情景泛化，例如动物园场景在预训练中大量出现，测试时在学校场景测试

评估方式：既看答案也看过程

模型：Qwen2.5架构，100M参数量

训练数据：30B数学题库划分为三份
pre-training：10B tokens，op=2~10
mid-training
post-training

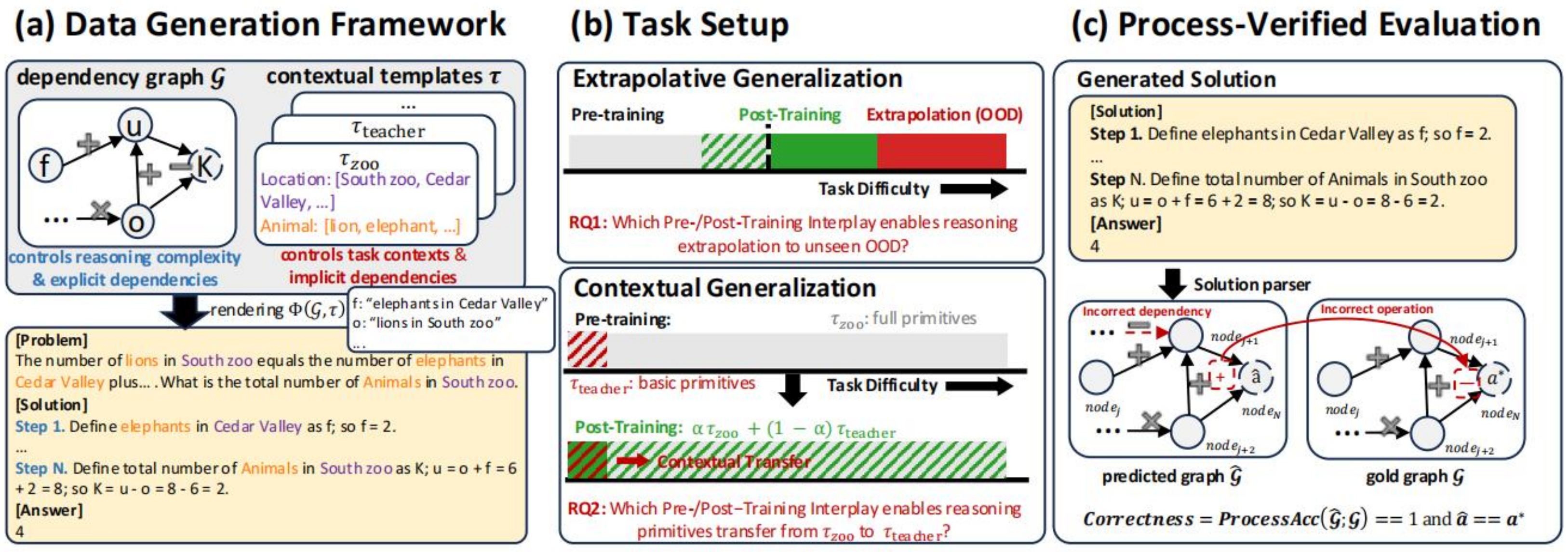

Figure 2: **Overview of the data generation framework, task setup, and process-verified evaluation.** The figure depicts the dependency graph $\mathcal{G}$ and contextual templates τ, the task setup for extrapolative and contextual generalization, and the process-verified evaluation framework that checks for correctness of reasoning steps.

研究1：RL何时能拓展base模型能力边界？

训练方案：
pre-training：固定10B，op=2~10
post-training：GRPO，200K，分别来自四个难度 op=7~10(ID)、op=9~12(mixed)、op=11~14(edge)、op=17~20(hard)

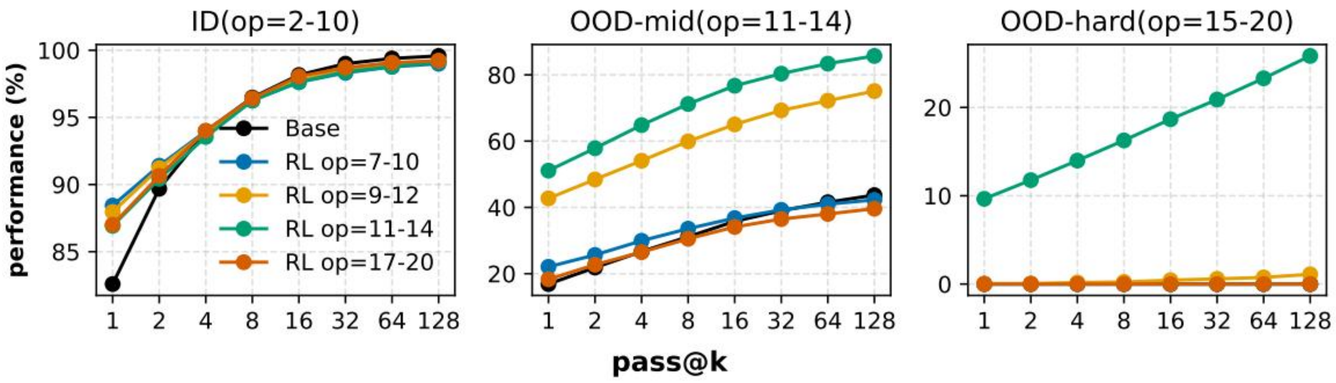

Figure 3: pass@k performance on three tasks: ID (op=2-10), OOD-edge (op=11-14), OOD-hard (op= (15-20)). RL is applied to four different data regimes (colors). RL on ID tasks never improves beyond the base model at pass@128. RL consistently improves pass@128 on harder tasks when applied beyond the base model's capacity.

结论：
1）对于base模型已经能解决的ID任务，RL无法带来能力上限提升。
2）对于OOD任务，当使用难度适中的数据进行RL，能提升base能力上限，太难/太简单都不行。
启发：RL训练数据需要精心构造，重点关注pass@1失败而pass@k成功的数据，难度应位于base模型的边界（edge）

研究2：预训练如何影响RL泛化?

训练方案:
pre-training：10B，op=2~20的情景A数据，少量op=2的情景B数据。控制A和B的不同比例。
post-training：固定，200K，50%A+50%B，op=2~20.

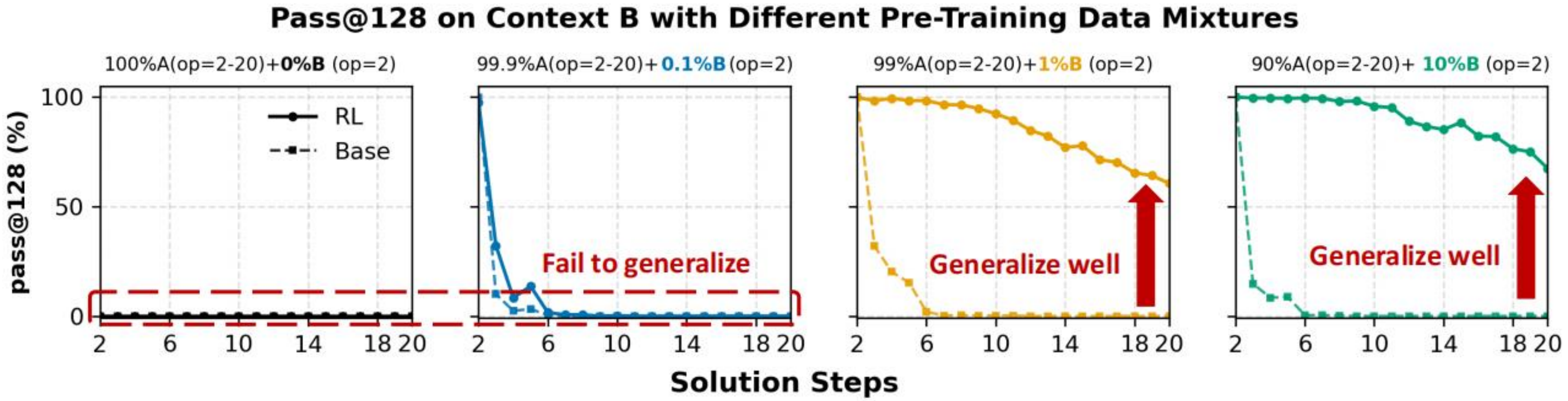

Figure 4: pass@128 performance on *context B* after post-trained with a 50% *context A* + 50% *context B* mixture. Different lines represent levels of pre-training exposure to long-tailed *context B* atomic op=2 examples. RL incentivizes contextual generalization when the model has minimal exposure ($\geq$1%) to *context B* in pre-training.

结论:
1）预训练数据B的含量过少或不包含，RL即使包含B，也无法实现情景泛化。
2）预训练数据中只需要极少比例的B（1%），RL就能实现情景泛化，即使对于op=20亦是如此。

研究2：预训练如何影响RL泛化？

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对于B的泛化，推理模式（运算图）是复用A的还是创造新的？
1）对于低难度任务，模型倾向于复用A的图结构
2）对于高难度任务，模型生成了更多新颖的推理图。

启发：预训练数据中应优先确保数据的广泛覆盖，作为seed用于post-training的能力激发

Replication or Creation? We examine the distribution of topological similarity between the generated correct *context B* graphs and the ground-truth topology from *context A* in Figure 5. High similarity indicates that the model primarily replicates existing *context A* reasoning patterns, while low similarity suggests the emergence of novel reasoning structures distinct from *context A*.

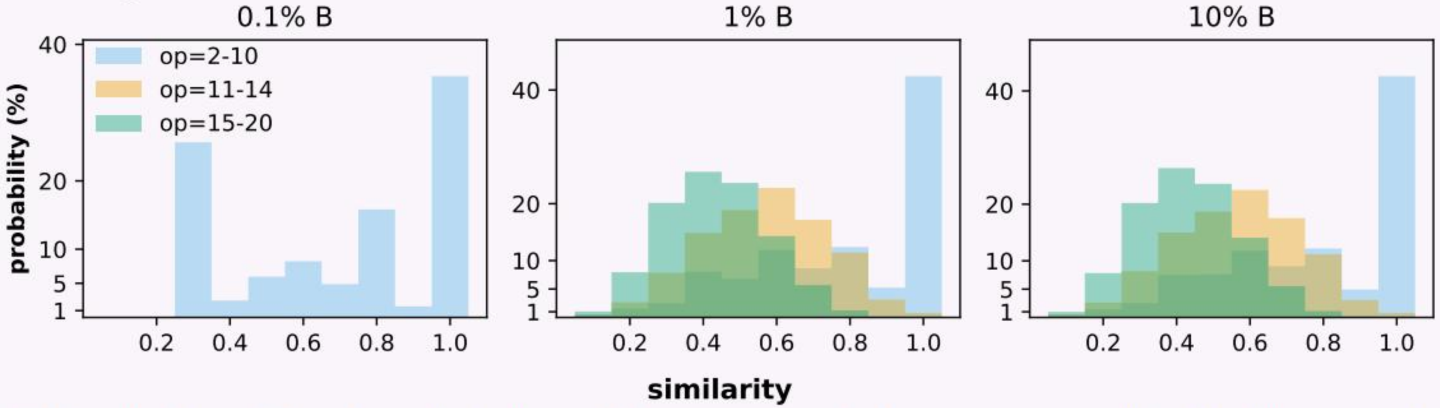

Figure 5: Distribution of topological similarity between generated correct *context B* and gold *context A* graphs.

We observe effects between task difficulty and pre-training exposure: 1) For simpler compositions (op=2-10), models tend to replicate existing patterns from *context A*. 2) As task complexity increases (op=11-20), models generate more novel structures, especially when pre-trained with sufficient exposure to *context B*.

研究3：mid-training与post-training如何相互影响?

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训练方案：固定pre-training（10B，op=2~10)；固定总“计算预算”，按不同比例分配给mid-training和post-training，训练数据为op=11~14的edge数据。full-mid-training使用1B，full RL使用100K。

Compute Budget Formulation. For fair comparison, we normalize both phases to *equivalent training tokens* based on flops. For mid-training, the consumption T_{mid} is the number of supervised tokens processed. For RL, the token-equivalent cost is approximated as:

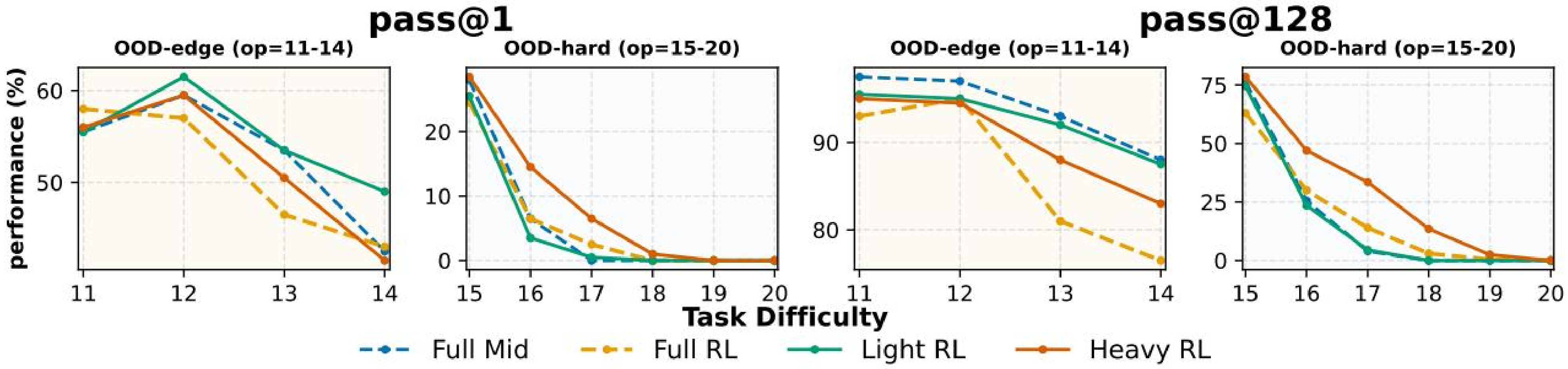
$$T_{\text{RL}} \approx \frac{5}{3} N \cdot r \cdot L_{\text{total}}, \tag{1}$$

where N is the number of RL samples, $r = 6$ the rollout multiplicity, and $L_{\text{total}=} = 2048$ the total token length[4].

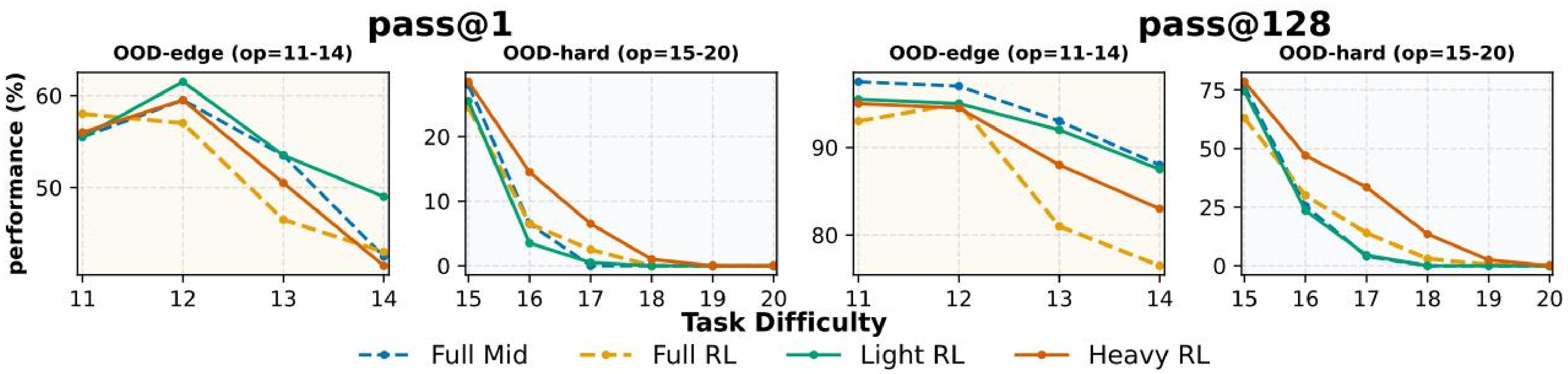
We systematically vary the RL allocation ratio $\beta \in [0, 1]$ to distribute the **total budget** T between the two phases:

$$T_{\text{mid}} = (1 - \beta) \cdot T, \quad T_{\text{RL}} = \beta \cdot T. \tag{2}$$

Task Setting. In this section, we explore the performance of five training configurations using the same base model pre-trained on 10B `op=2-10` data: **Full mid-training** on 1B supervised tokens from the `op=11-14` range, **Full RL** with 100 steps of batch size 1024 from the same `op=11-14` range, and three mixing strategies—*Light-RL* ($\beta = 0.2$), *Medium-RL* ($\beta = 0.5$), and *Heavy-RL* ($\beta = 0.8$), which balance mid-training and RL under an equivalent compute budget. The compute budget formulation in Section 5 allows for a direct comparison of data mixture strategies. Detailed training setup can be found in Appendix A.10.



研究3：mid-training与post-training如何相互影响?



结论：

1）对于OOD-edge（训练数据覆盖），Ligth RL表现最佳。即在已知难度上表现好，mid-training很重要。

2）对于OOD-hard（训练数据不包含），需要RL的探索拓展能力边界

启发：需要根据任务目标难度合理分配算力

近期研究发现Qwen模型对于zero RL的训练结果远优于llama等模型，一个原因可能是Qwen的mid-training阶段使得其本身就有一定的推理能力，缩小了预训练与RL的差距。

研究4：过程级奖励有助于生成正确的CoT

训练方案：使用op=11~14数据进行RL。

仅使用结果奖励（RLVR），CoT过程可能是错误的。研究过程级奖励对RL的影响

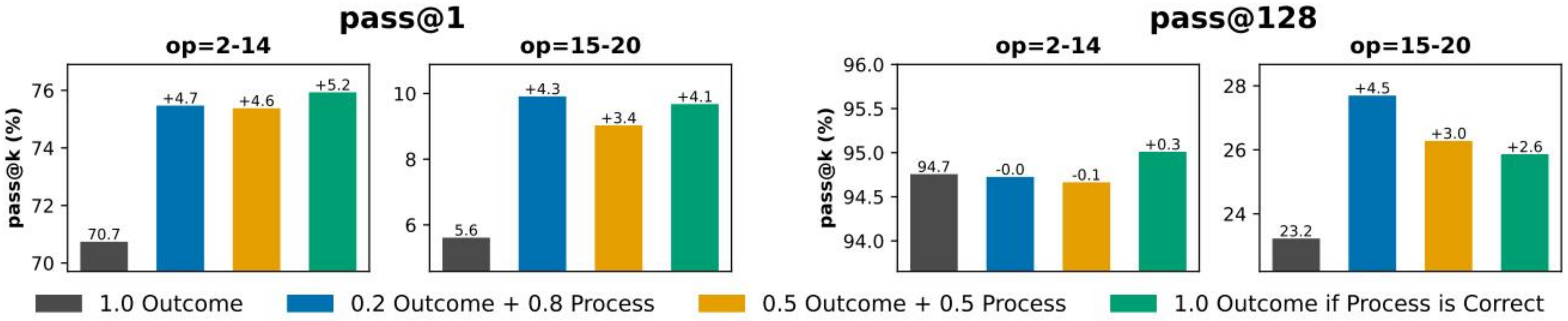


Figure 7: pa
mixing strate
performance
reward-
surable

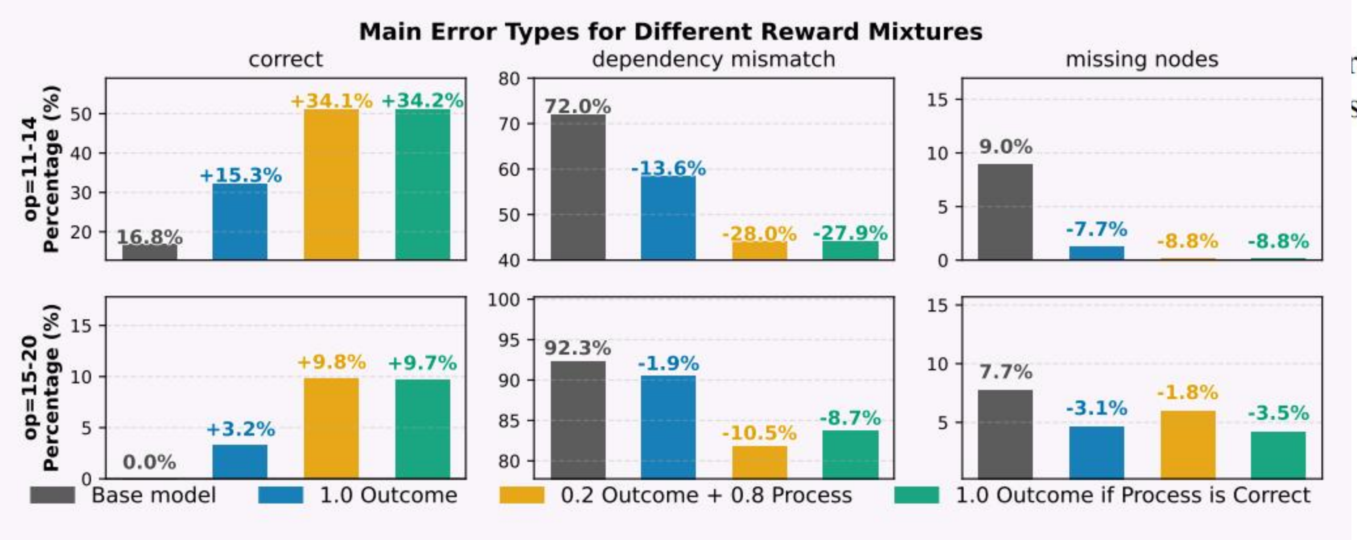


Figure 8: Effects of reward mixtures on reasoning correctness and structural error types.

结果：过程奖励显著提升了性能，尤其是在pass@1上提升较大；过程奖励对于泛化亦十分重要

总结

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1. 预训练的为 RL 打基础，决定RL扩展潜力

RL仅在预训练留有足够提升空间（任务未被过度训练）时产生扩展；否则对于覆盖相对饱和的任务，仅优化现有分布（即 pass@K -> pass@1 转移）。

启示：预训练决定上限，RL无法从零发明技能，仅组合现有原语。

2. mid-training的能够桥接预训练分布与RL分布，分配适当用于给mid-training能提升综合能力

3. RL需要精心选择edge数据，并设计合适的目标函数（即包含过程奖励也包含结果奖励）